VIME

(Variational Information Maximizing Exploration)

VIME is an exploration strategy

- Scalable and effective exploration is a key challenge.
- Most RL relies on simple heuristics such as ε-greedy or Gaussian noise.
- VIME maximizes information gain about the agent's belief of environment dynamics.
- VIME achieves significant performance improvement in variety of continuous control tasks and algorithms, including tasks with very sparse rewards.

Methodology

Curiosity-driven exploration

Agent models the environment dynamics $p(s_{t+1}|s_t, a_t; \theta)$

Agents choose actions that cause large updates to the dynamics model.

The agent should take actions that maximize the reduction in uncertainty about the dynamics.

$$\sum_t \left(H(\Theta|\xi_t, a_t) - H(\Theta|S_{t+1}, \xi_t, a_t)\right)$$

The individual terms equal the mutual information.

$$I\left(S_{t+1}; \Theta|\xi_t, a_t\right) = \mathbb{E}_{s_{t+1}\sim\mathcal{P}(\cdot|\xi_t, a_t)}\left[D_{\mathrm{KL}}[p(\theta|\xi_t, a_t, s_{t+1}) \,\|\, p(\theta|\xi_t)]\right]$$

(information gain)

Therefore, the agent is encouraged to actions that lead to states that are maximally informative about the dynamics model. The KL divergence can be interpreted as *information gain*.

The trade-off between exploitation and exploration can now be realized explicitly using information gain as follows:

$$r'(s_t, a_t, s_{t+1}) = r(s_t, a_t) + \eta D_{\mathrm{KL}}[p(\theta|\xi_t, a_t, s_{t+1}) \,\|\, p(\theta|\xi_t)]$$

Where η is a hyperparameter controlling the urge to explore.

Biggest practical issue:
Calculating the posterior $p(\theta|\xi_t, a_t, s_{t+1})$ is generally intractable.

By Bayes' rule,

$$p(\theta|\xi_t, a_t, s_{t+1}) = \frac{p(\theta|\xi_t)p(s_{t+1}|\xi_t, a_t; \theta)}{p(s_{t+1}|\xi_t, a_t)}$$

Where

$$p(s_{t+1}|\xi_t, a_t) = \int_{\Theta} p(s_{t+1}|\xi_t, a_t; \theta)p(\theta|\xi_t)d\theta$$

In general, the integral tends to be intractable when using highly expressive parametrized models such as NNs.

Variational Bayes

Solution:
Approximate information gain by variational inference method

Calculating the posterior $p(\theta|\mathcal{D})$ is intractable. Instead, we approximate it through an alternative distribution $q(\theta;\phi)$.

In other words, minimize $D_{\mathrm{KL}}[q(\theta;\phi)\,\|\,p(\theta|\mathcal{D})]$

Approximation is done by maximizing *variational lower bound* –

$$L[q(\theta;\phi),\mathcal{D}] = \mathbb{E}_{\theta\sim q(\cdot;\phi)}\left[\log p(\mathcal{D}|\theta)\right] - D_{\mathrm{KL}}[q(\theta;\phi)\,\|\,p(\theta)]$$

Rather than computing information gain explicitly, we compute an approximation to it, leading to the following total reward:

$$r'(s_t,a_t,s_{t+1}) = r(s_t,a_t) + \eta D_{\mathrm{KL}}[q(\theta;\phi_{t+1})\,\|\,q(\theta;\phi_t)]$$

Where ϕ_{t+1} is the updated parameters and ϕ_t is the old parameters representing the agent's belief.

Implementation

We utilize a BNN for parametrizing the agent's dynamics model. The weight distribution is given by the fully factorized Gaussian distribution:

$$q(\theta;\phi) = \prod_{i=1}^{|\Theta|} \mathcal{N}(\theta_i|\mu_i;\sigma_i^2)$$

hence, $\phi = \{\mu, \sigma\}$

The parameter ϕ can be computed by following minimization:

$$\phi' = \arg\min_{\phi} \Big[\overbrace{\underbrace{D_{\mathrm{KL}}[q(\theta;\phi)\,\|\,q(\theta;\phi_{t-1})]}_{\ell_{\mathrm{KL}}(q(\theta;\phi))} - \mathbb{E}_{\theta\sim q(\cdot;\phi)}\left[\log p(s_t|\xi_t, a_t;\theta)\right]}^{\ell(q(\theta;\phi),s_t)} \Big]$$

The second term is approximated through sampling N samples:

$$\mathbb{E}_{\theta\sim q(\cdot;\phi)}\left[\log p(\mathcal{D}|\theta)\right] \approx \frac{1}{N}\sum_{i=1}^{N} \log p(\mathcal{D}|\theta_i)$$

Algorithm

Algorithm 1: Variational Information Maximizing Exploration (VIME)

for each epoch n **do**

  for each timestep t in each trajectory generated during n **do**

    Generate action $a_t \sim \pi_\alpha(s_t)$ and sample state $s_{t+1} \sim \mathcal{P}(\cdot|\xi_t, a_t)$, get $r(s_t, a_t)$.

    Add triplet (s_t, a_t, s_{t+1}) to FIFO replay pool $\mathcal{R}$.

    Compute $D_{\mathrm{KL}}[q(\theta; \phi'_{n+1}) \,\|\, q(\theta; \phi_{n+1})]$ by approximation $\nabla^\top H^{-1} \nabla$, following Eq. (16) for diagonal BNNs, or by optimizing Eq. (12) to obtain ϕ'_{n+1} for general BNNs.

    Divide $D_{\mathrm{KL}}[q(\theta; \phi'_{n+1}) \,\|\, q(\theta; \phi_{n+1})]$ by median of previous KL divergences.

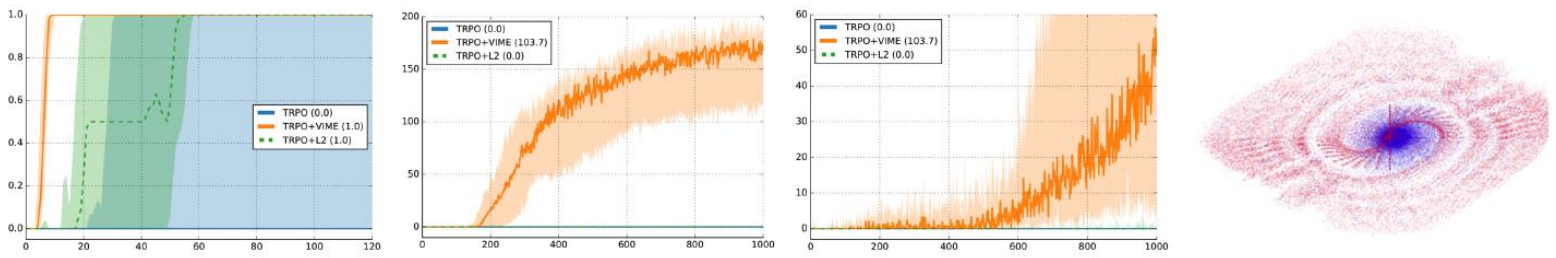
    Construct $r'(s_t, a_t, s_{t+1}) \leftarrow r(s_t, a_t) + \eta D_{\mathrm{KL}}[q(\theta; \phi'_{n+1}) \,\|\, q(\theta; \phi_{n+1})]$, following Eq. (7).

  Minimize $D_{\mathrm{KL}}[q(\theta; \phi_n) \,\|\, p(\theta)] - \mathbb{E}_{\theta \sim q(\cdot; \phi_n)} [\log p(\mathcal{D}|\theta)]$ following Eq. (6), with $\mathcal{D}$ sampled randomly from $\mathcal{R}$, leading to updated posterior $q(\theta; \phi_{n+1})$.

  Use rewards $\{r'(s_t, a_t, s_{t+1})\}$ to update policy π_α using any standard RL method.

Experiments

Tasks with sparse rewards

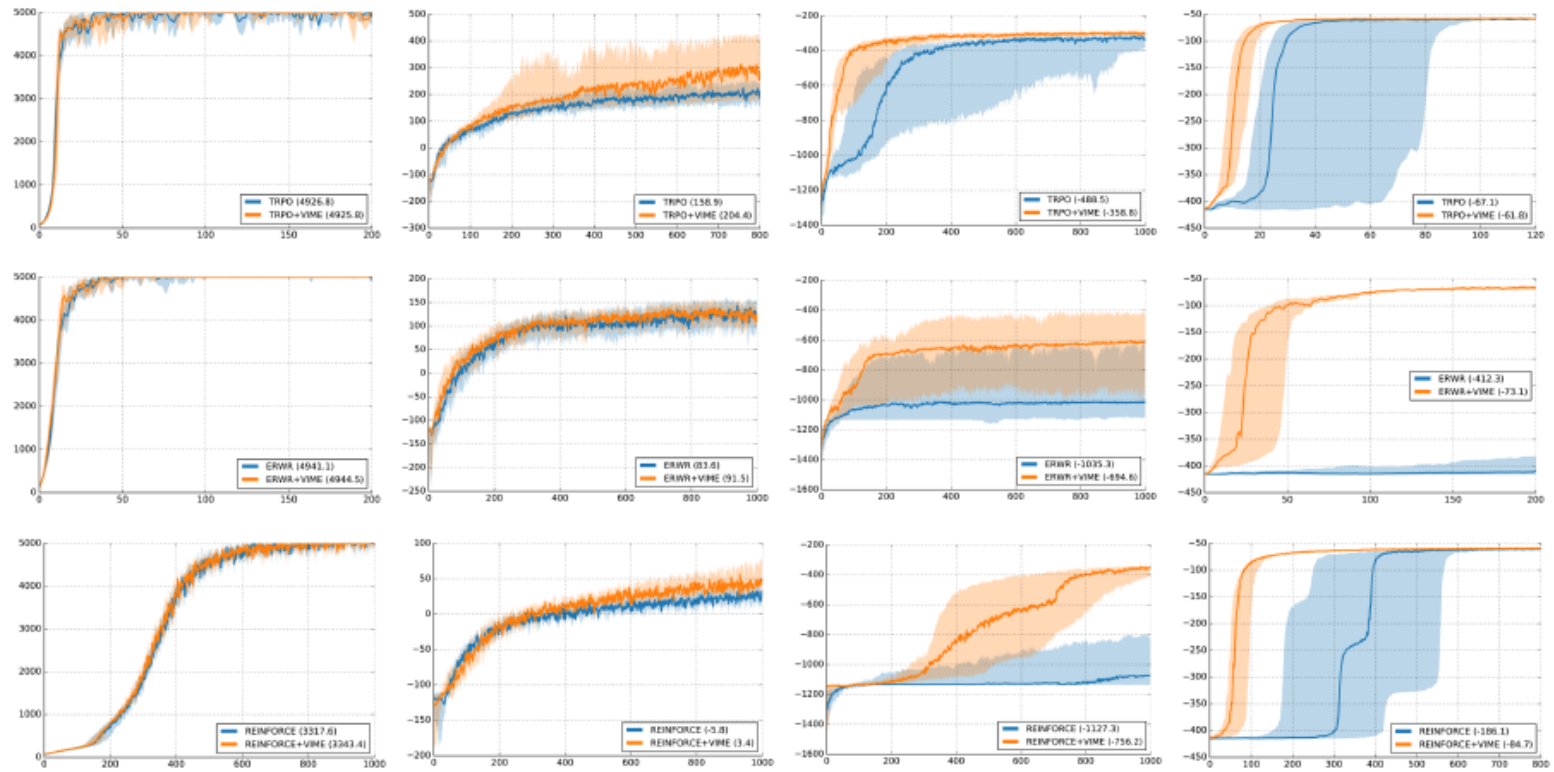


(a) MountainCar (b) CartPoleSwingup (c) HalfCheetah (d) state space

Figure 1: (a,b,c) TRPO+VIME versus TRPO on tasks with sparse rewards; (d) comparison of TRPO+VIME (red) and TRPO (blue) on MountainCar: visited states until convergence

- Domains with sparse rewards are difficult to solve through naïve exploration because it lacks a signal on how to improve policy.
- This experiment demonstrates that curiosity drives the agent to explore, even in the absence of any initial reward.

Paired with different RL methods



(a) CartPole (b) CartPoleSwingup (c) DoublePendulum (d) MountainCar

Figure 2: Performance of TRPO (top row), ERWR (middle row), and REINFORCE (bottom row) with (+VIME) and without exploration for different continuous control tasks.

How η trades off XX behavior

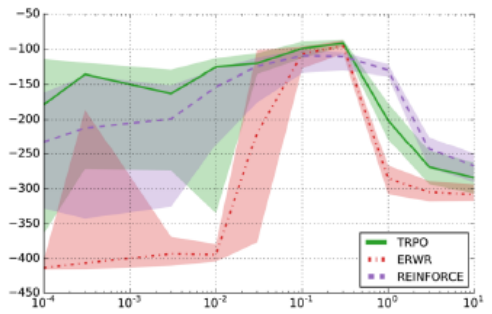


Figure 4: VIME: performance over the first few iterations for TRPO, REINFORCE, and ERWR i.f.o. η on MountainCar.

- Too high η prioritizes exploration over getting more reward
- Too low η reduces the method to the baseline algorithm as the intrinsic reward becomes negligible.
- There is a wide range of η for which the task is best solved, across different algorithms.